

POCKETSMART AI: SYSTEM REQUIREMENTS SPECIFICATION

1. Functional Requirements (FR)

- **FR1: User Authentication & Security:** The system must permit dynamic user onboarding (/register) and secure sign-in (/login) utilizing OAuth2 JWT bearer tokens.
- **FR2: Modular Budget Planning:**
 - **FR2.1 (Home Planner):** Must accept monetary constraints, room flags, and counts for electrical fixtures, returning a 25-25-50 split (Lighting, Cooling, Furniture).
 - **FR2.2 (Party Planner):** Must accept guest counts, event classification, and catering flags, generating a 30-50-20 split (Venue, Food, Decor).
 - **FR2.3 (Jewelry Planner):** Must accept occasion types and lifestyle preferences, allocating a 60-40 split between primary statement pieces and secondary accessory sets.
- **FR3: Multimodal Vision Processing:** The system must support multipart image uploads (multipart/form-data) and process image streams alongside prompt text using Gemini Generative AI models.
- **FR4: Deterministic Fallback Mechanism:** In cases of API throttling, rate limiting, or connection latency, the backend must return a calibrated Indian market budget breakdown without throwing a 500 Internal Server Error.
- **FR5: Automated E-Commerce Deep-Linking:** All suggested items must be parsed with `urllib.parse` to construct live search query links to Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, and Tanishq.

2. Non-Functional Requirements (NFR)

- **NFR1: Performance & Latency:** Budget breakdown responses must be returned to the client in less than 1.5 seconds under normal API conditions, and under 50 milliseconds when served via the local fallback engine.

- **NFR2: High Availability & Fault Tolerance:** The application must maintain 99.9% availability through isolated exception handling in `services/gemini_utils.py` and `routes/planners.py`.
- **NFR3: Security & Cryptography:** Passwords must be hashed using industry-standard Bcrypt algorithms. Access tokens must be signed with secret keys (HS256) and expire within 60 minutes.
- **NFR4: Responsive Cross-Platform UI:** The frontend must adapt seamlessly across desktop monitors, tablets, and smartphones using Bootstrap 5 grid utilities.